Predicting Open-Ended Response

Group 15

Contents

Question

> **What predicts having any likes/dislikes for either party?** Who are we including/excluding in the open-ended sample?

A respondent is **excluded** from the open-ended sample if either

- they answered "No" to the like/dislike screener (`*_binary == 5`), **or**
- they said "Yes" but provided no actual open-ended text (`*_binary == 1` *and* the text field is `NA`).

1 Data prep

```r
setwd("/Users/yzhang/Desktop/r_projects/mdm_final")
rm(list = ls())
pacman::p_load(glmnet, coefplot, tidyverse, sandwich, lmtest)

data <- read.csv("data.csv", header = TRUE, na.strings = c("", "NA"))
dim(data)
```

1.1 Recode the four outcomes

For each like/dislike pair, the new 0/1 outcome is:

- **1** if `*_binary == 1` AND the text field is non-empty (gave usable text)
- **0** if `*_binary == 5` OR (`*_binary == 1` AND text is NA)
- **NA** if `*_binary` itself is NA (drop these rows later)

```r
recode_y <- function(bin, text) {
  text_present <- !is.na(text) & nchar(trimws(text)) > 0
  y <- rep(NA_integer_, length(bin))
  y[bin == 5]                  <- 0L
  y[bin == 1 & !text_present]  <- 0L
  y[bin == 1 &  text_present]  <- 1L
  y
}

data$y_dem_like    <- recode_y(data$dem_like_binary,    data$dem_like)
data$y_dem_dislike <- recode_y(data$dem_dislike_binary, data$dem_dislike)
data$y_rep_like    <- recode_y(data$rep_like_binary,    data$rep_like)
data$y_rep_dislike <- recode_y(data$rep_dislike_binary, data$rep_dislike)

sapply(data[, c("y_dem_like","y_dem_dislike","y_rep_like","y_rep_dislike")],
       table, useNA = "ifany")
```

1.2 Drop columns that should never be predictors

We drop:

- IDs and the survey weight (not substantive predictors).
- The raw outcome components.
- `age_cohort` (collinear with `age`).
- Items not asked every year (`understand_issue`, `too_complicated`, `power_matters`, `dem_satis`) and items with high missingness (`dem_liking_cses`, `rep_liking_cses`, `therm_lib`, `therm_consv`). Keeping them would drop the analytic sample below 15,000 after `na.omit`.

```r
drop_cols <- c("caseid", "uniq_id", "weight",
               "dem_like_binary", "dem_dislike_binary",
               "rep_like_binary", "rep_dislike_binary",
               "dem_like", "dem_dislike", "rep_like", "rep_dislike",
```

```
                "age_cohort",
                "understand_issue", "too_complicated",
                "power_matters", "dem_satis",
                "dem_liking_cses", "rep_liking_cses",
                "therm_lib", "therm_consv")
data <- data[, !(names(data) %in% drop_cols)]
```

1.3 Collapse multi-level ordinals

Per-level coefficients on 7-point scales are hard to read in a table, so we collapse them. Out-of-range codes (DK / RF / 0 / 9) fall through to `NA` and get dropped by `na.omit` later. The middle category is set as the reference so coefficients read as "vs moderates / independents."

```
collapse_7pt <- function(x, labels) {
  factor(dplyr::case_when(
    x %in% 1:3 ~ labels[1],
    x == 4     ~ labels[2],
    x %in% 5:7 ~ labels[3],
    TRUE       ~ NA_character_
  ), levels = c(labels[2], labels[1], labels[3]))   # middle first = reference
}

data$pid7     <- collapse_7pt(data$pid7,     c("Dem", "Ind", "Rep"))
data$ideology <- collapse_7pt(data$ideology, c("Lib", "Mod", "Con"))
data$dem_ideo <- collapse_7pt(data$dem_ideo, c("Lib", "Mod", "Con"))
data$rep_ideo <- collapse_7pt(data$rep_ideo, c("Lib", "Mod", "Con"))

data$educ7 <- factor(dplyr::case_when(
  data$educ7 %in% 1:2 ~ "NoHS",
  data$educ7 %in% 3:5 ~ "HS_SomeCol",
  data$educ7 %in% 6:7 ~ "BA_plus",
  TRUE                ~ NA_character_
), levels = c("HS_SomeCol", "NoHS", "BA_plus"))   # HS = reference
```

1.4 Make year a factor and convert remaining categoricals

```
data$year <- factor(data$year)

cont_vars <- c("age", "dem_therm", "rep_therm")
y_vars    <- c("y_dem_like", "y_dem_dislike", "y_rep_like", "y_rep_dislike")
cat_vars  <- setdiff(names(data), c(cont_vars, y_vars, "year"))

# already-factor columns pass through unchanged; numeric ones get factored
data[cat_vars] <- lapply(data[cat_vars],
                         function(z) if (is.factor(z)) z else factor(z))
```

1.5 Listwise-delete missing predictors

```r
data_full <- na.omit(data)
dim(data_full)
```

```
## [1] 17844    25
```

```r
# Every outcome should have both 0s and 1s, every year should have substantial counts.
cat("\nClass distribution of each outcome:\n")
```

```
##
## Class distribution of each outcome:
```

```r
for (y in y_vars) {
  cat(sprintf("  %-15s ", y)); print(table(data_full[[y]]))
}
```

```
##   y_dem_like
##    0    1
## 8388 9456
##   y_dem_dislike
##     0     1
##  7548 10296
##   y_rep_like
##    0    1
## 9601 8243
##   y_rep_dislike
##     0     1
##  6707 11137
```

```r
cat("\nYear counts:\n"); print(table(data_full$year))
```

```
##
## Year counts:
```

```
##
## 2008 2012 2016 2020 2024
## 1251 4456 2618 5772 3747
```

1.6 Build X and Y

```
Y_dl <- data_full$y_dem_like

# RHS = everything except the four outcomes
rhs  <- setdiff(names(data_full), y_vars)
form <- as.formula(paste("~", paste(rhs, collapse = " + ")))
X    <- model.matrix(form, data = data_full)[, -1]   # drop intercept col
dim(X)
```

```
## [1] 17844    50
```

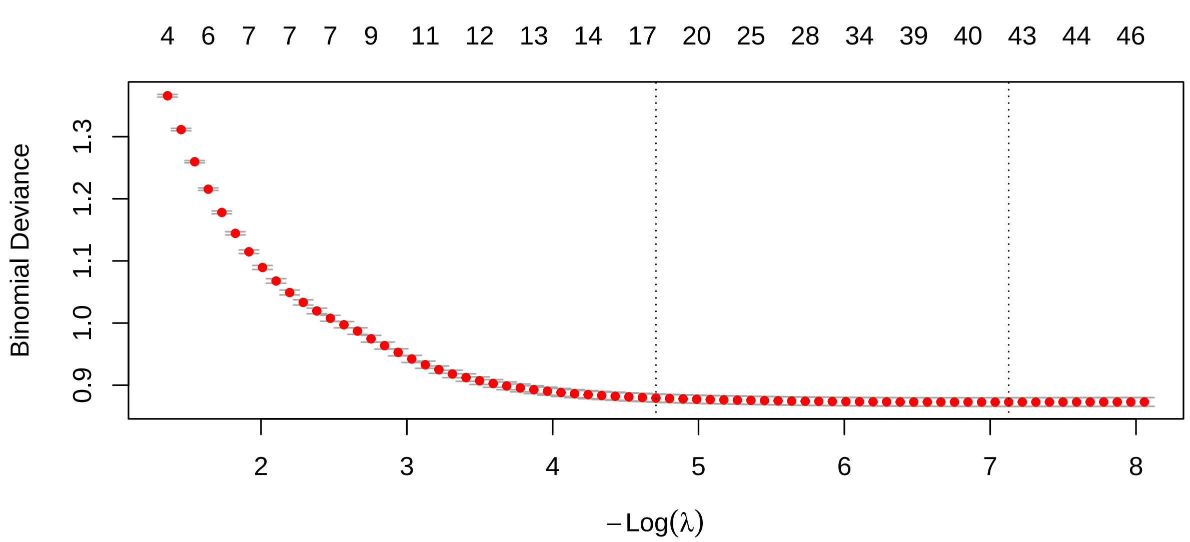
1.7 Force year in via penalty.factor

```
pf <- ifelse(grepl("^year", colnames(X)), 0, 1)
sum(pf == 0)   # number of year dummies forced in (= 4 for 5 years)
```

```
## [1] 4
```

1.8 Run cv.glmnet

```
set.seed(10)
fit.dl.cv <- cv.glmnet(X, Y_dl,
                       alpha          = 1,
                       family         = "binomial",
                       nfolds         = 10,
                       penalty.factor = pf)
plot(fit.dl.cv)
```



```r
fit.dl.cv$lambda.min
```

```
## [1] 0.000803
```

```r
fit.dl.cv$lambda.1se
```

```
## [1] 0.00902
```

1.9 Extract selected variables at `lambda.1se`

```r
coef.dl <- extract.coef(fit.dl.cv, lambda = fit.dl.cv$lambda.1se)
coef.dl
```

```
##                         Value SE       Coefficient
## (Intercept)         -1.80e+00 NA       (Intercept)
## year2012            -5.33e-01 NA          year2012
## year2016            -1.30e+00 NA          year2016
## year2020            -4.89e-01 NA          year2020
## year2024            -2.54e-01 NA          year2024
## dem_therm            3.02e-02 NA         dem_therm
## rep_therm           -1.52e-02 NA         rep_therm
## pid7Dem              5.76e-01 NA           pid7Dem
## party_diff2          5.20e-01 NA       party_diff2
## party_more_consv2    4.56e-02 NA party_more_consv2
## party_more_consv9   -2.24e-01 NA party_more_consv9
## dem_ideoLib          1.43e-01 NA       dem_ideoLib
## rep_ideoCon          6.73e-01 NA       rep_ideoCon
## right_track1         4.32e-02 NA      right_track1
## right_track2        -5.65e-06 NA      right_track2
## ideologyLib          2.63e-01 NA       ideologyLib
## educ7BA_plus         4.56e-01 NA      educ7BA_plus
## health_insur2       -4.32e-02 NA     health_insur2
```

```r
var.dl  <- rownames(coef.dl)[-1]    # column names from model.matrix
```

1.10 Refit with `glm` (relaxed LASSO) and apply clustered SEs

`model.matrix` produces names like `pid7Rep`, `race_eth5`. We map them back to parent variable names to write the `glm` formula.

```r
parent_var <- function(cols, vars) {
  hits <- sapply(cols, function(cn) {
    m <- vars[startsWith(cn, vars)]
    if (length(m)) m[which.max(nchar(m))] else NA_character_
  })
  unique(na.omit(hits))
}
```

```r
vars.dl <- parent_var(var.dl, names(data_full))
vars.dl <- union(vars.dl, "year")   # year is forced in regardless

f.dl <- as.formula(paste("y_dem_like ~", paste(vars.dl, collapse = " + ")))
fit.dl <- glm(f.dl, data = data_full, family = binomial)

V.dl <- sandwich::vcovCL(fit.dl, cluster = ~ year, type = "HC1")
coeftest(fit.dl, vcov. = V.dl)
```

```
##
## z test of coefficients:
##
##                     Estimate Std. Error z value Pr(>|z|)
## (Intercept)         -2.32547    0.14781  -15.73  < 2e-16 ***
## year2012            -0.60169    0.03947  -15.25  < 2e-16 ***
## year2016            -1.44903    0.05337  -27.15  < 2e-16 ***
## year2020            -0.58659    0.03548  -16.53  < 2e-16 ***
## year2024            -0.30005    0.02951  -10.17  < 2e-16 ***
## dem_therm            0.03384    0.00134   25.16  < 2e-16 ***
## rep_therm           -0.01808    0.00122  -14.88  < 2e-16 ***
## pid7Dem              0.69897    0.07235    9.66  < 2e-16 ***
## pid7Rep              0.21935    0.12593    1.74   0.0815 .
## party_diff2          0.59327    0.06933    8.56  < 2e-16 ***
## party_diff9         -0.13495    0.51817   -0.26   0.7945
## party_more_consv2    0.11896    0.08296    1.43   0.1516
## party_more_consv9 -0.26555    0.12174   -2.18   0.0292 *
## dem_ideoLib          0.37961    0.04238    8.96  < 2e-16 ***
## dem_ideoCon          0.28050    0.10328    2.72   0.0066 **
## rep_ideoLib          0.30038    0.12632    2.38   0.0174 *
## rep_ideoCon          0.90041    0.09214    9.77  < 2e-16 ***
## right_track2        -0.19648    0.14585   -1.35   0.1779
## ideologyLib          0.32427    0.05749    5.64  1.7e-08 ***
## ideologyCon          0.01957    0.07226    0.27   0.7866
## educ7NoHS           -0.17245    0.08908   -1.94   0.0529 .
## educ7BA_plus         0.53561    0.09942    5.39  7.2e-08 ***
## health_insur2       -0.22428    0.08000   -2.80   0.0051 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

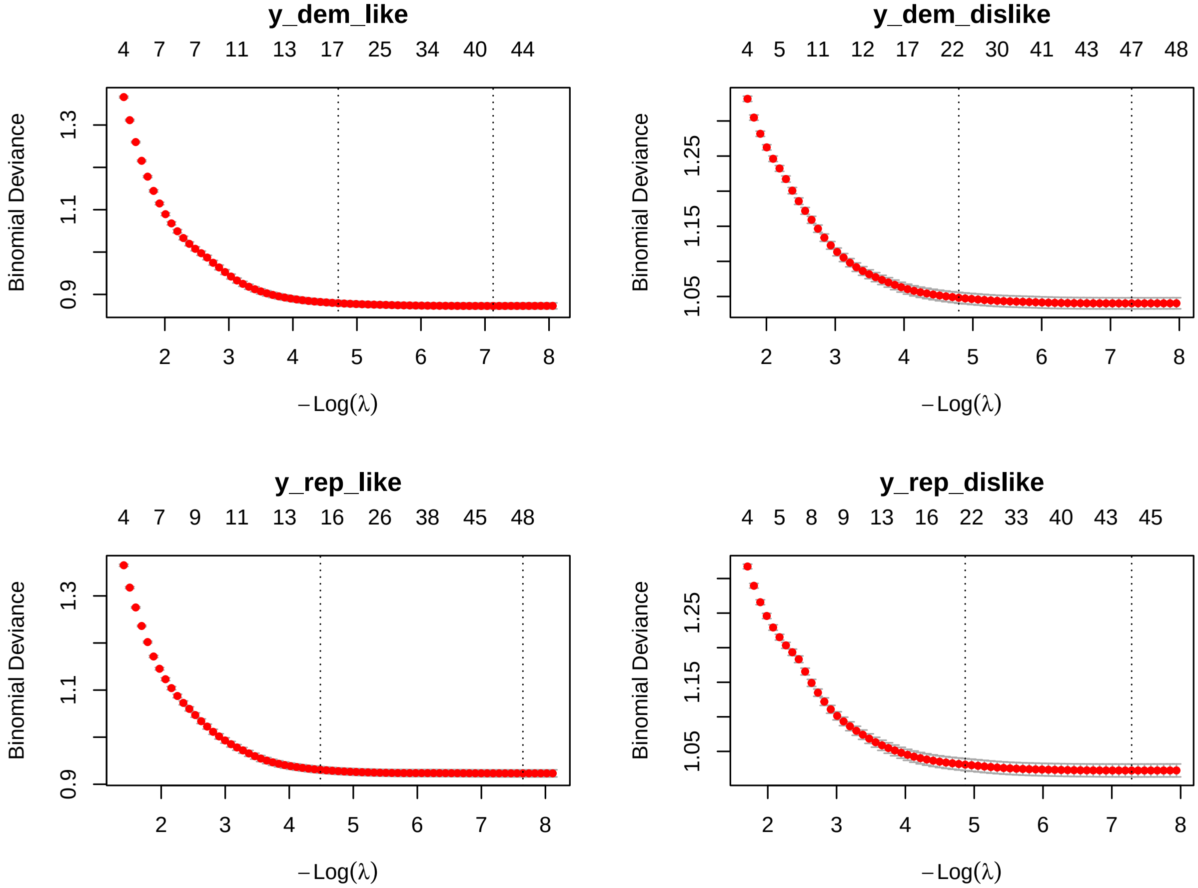
2 Loop over all four outcomes

Same logic, wrapped in a function.

```r
run_one <- function(y_name, data_full, seed = 10) {
  Y    <- as.numeric(data_full[[y_name]])
  stopifnot(length(unique(Y)) == 2)   # fail loud if Y is constant

  rhs  <- setdiff(names(data_full), y_vars)
  form <- as.formula(paste("~", paste(rhs, collapse = " + ")))
  X    <- model.matrix(form, data = data_full)[, -1]
  pf   <- ifelse(grepl("^year", colnames(X)), 0, 1)

  set.seed(seed)
  cv <- cv.glmnet(X, Y, alpha = 1, family = "binomial",
                  nfolds = 10, penalty.factor = pf)

  coef.tab <- extract.coef(cv, lambda = cv$lambda.1se)
  vars     <- parent_var(rownames(coef.tab)[-1], names(data_full))
  vars     <- union(vars, "year")

  f   <- as.formula(paste(y_name, "~", paste(vars, collapse = " + ")))
  fit <- glm(f, data = data_full, family = binomial)
  V   <- sandwich::vcovCL(fit, cluster = ~ year, type = "HC1")

  list(cv = cv, fit = fit, vcov = V,
       coef_test  = coeftest(fit, vcov. = V),
       n          = nrow(data_full),
       lambda_1se = cv$lambda.1se,
       n_selected = nrow(coef.tab) - 1)
}

results <- lapply(y_vars, run_one, data_full = data_full)
names(results) <- y_vars

data.frame(outcome    = y_vars,
           n          = sapply(results, `[[`, "n"),
           lambda_1se = sapply(results, `[[`, "lambda_1se"),
           n_selected = sapply(results, `[[`, "n_selected"))
```

```
##                     outcome     n lambda_1se n_selected
## y_dem_like       y_dem_like 17844    0.00902         17
## y_dem_dislike y_dem_dislike 17844    0.00827         23
## y_rep_like       y_rep_like 17844    0.01127         15
## y_rep_dislike y_rep_dislike 17844    0.00768         20
```

```r
par(mfrow = c(2, 2))
for (y in y_vars) { plot(results[[y]]$cv); title(y, line = 2.5) }
```




```r
par(mfrow = c(1, 1))
```

```r
for (y in y_vars) {
  cat("\n----- ", y, " -----\n")
  print(results[[y]]$coef_test)
}
```

```
##
## -----  y_dem_like  -----
##
## z test of coefficients:
##
##                     Estimate Std. Error z value Pr(>|z|)
## (Intercept)         -2.32547    0.14781  -15.73  < 2e-16 ***
## year2012            -0.60169    0.03947  -15.25  < 2e-16 ***
## year2016            -1.44903    0.05337  -27.15  < 2e-16 ***
## year2020            -0.58659    0.03548  -16.53  < 2e-16 ***
## year2024            -0.30005    0.02951  -10.17  < 2e-16 ***
## dem_therm            0.03384    0.00134   25.16  < 2e-16 ***
## rep_therm           -0.01808    0.00122  -14.88  < 2e-16 ***
## pid7Dem              0.69897    0.07235    9.66  < 2e-16 ***
## pid7Rep              0.21935    0.12593    1.74   0.0815 .
## party_diff2          0.59327    0.06933    8.56  < 2e-16 ***
```

```
## party_diff9       -0.13495    0.51817   -0.26   0.7945
## party_more_consv2  0.11896    0.08296    1.43   0.1516
## party_more_consv9 -0.26555    0.12174   -2.18   0.0292 *
## dem_ideoLib        0.37961    0.04238    8.96  < 2e-16 ***
## dem_ideoCon        0.28050    0.10328    2.72   0.0066 **
## rep_ideoLib        0.30038    0.12632    2.38   0.0174 *
## rep_ideoCon        0.90041    0.09214    9.77  < 2e-16 ***
## right_track2      -0.19648    0.14585   -1.35   0.1779
## ideologyLib        0.32427    0.05749    5.64  1.7e-08 ***
## ideologyCon        0.01957    0.07226    0.27   0.7866
## educ7NoHS         -0.17245    0.08908   -1.94   0.0529 .
## educ7BA_plus       0.53561    0.09942    5.39  7.2e-08 ***
## health_insur2     -0.22428    0.08000   -2.80   0.0051 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## -----  y_dem_dislike  -----
##
## z test of coefficients:
##
##                   Estimate Std. Error z value Pr(>|z|)
## (Intercept)       -1.98234    0.25437   -7.79  6.5e-15 ***
## year2012           0.54411    0.03870   14.06  < 2e-16 ***
## year2016           0.05497    0.01644    3.34  0.00083 ***
## year2020           0.61980    0.03322   18.66  < 2e-16 ***
## year2024           0.73641    0.05051   14.58  < 2e-16 ***
## reg_census2       -0.14980    0.06856   -2.19  0.02888 *
## reg_census3       -0.01309    0.04276   -0.31  0.75949
## reg_census4        0.14784    0.01556    9.50  < 2e-16 ***
## dem_therm         -0.02730    0.00216  -12.61  < 2e-16 ***
## pid7Dem            0.26594    0.09651    2.76  0.00586 **
## pid7Rep            0.38206    0.06225    6.14  8.4e-10 ***
## party_diff2        0.54101    0.06199    8.73  < 2e-16 ***
## party_diff9        0.21566    0.55896    0.39  0.69962
## party_more_consv2  0.63295    0.09478    6.68  2.4e-11 ***
## party_more_consv9  0.13917    0.08831    1.58  0.11506
## dem_ideoLib        0.34370    0.05890    5.84  5.4e-09 ***
## dem_ideoCon        0.16140    0.04591    3.52  0.00044 ***
## rep_ideoLib        0.24023    0.05309    4.52  6.0e-06 ***
## rep_ideoCon        0.79580    0.09007    8.84  < 2e-16 ***
## right_track2       0.29181    0.13279    2.20  0.02798 *
## ideologyLib        0.16346    0.07138    2.29  0.02203 *
## ideologyCon        0.45452    0.07043    6.45  1.1e-10 ***
## age                0.00922    0.00147    6.25  4.0e-10 ***
## gender2           -0.28646    0.05917   -4.84  1.3e-06 ***
## gender3           -0.22712    0.05811   -3.91  9.3e-05 ***
## race_eth2         -0.15417    0.03375   -4.57  4.9e-06 ***
## race_eth3         -0.36577    0.07419   -4.93  8.2e-07 ***
## race_eth4         -0.14911    0.27584   -0.54  0.58881
## race_eth5         -0.22677    0.02699   -8.40  < 2e-16 ***
## race_eth6          0.14330    0.03949    3.63  0.00028 ***
## educ7NoHS         -0.46190    0.06262   -7.38  1.6e-13 ***
## educ7BA_plus       0.58788    0.04239   13.87  < 2e-16 ***
```

```
## health_insur2     -0.14210    0.09777   -1.45  0.14613
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## -----  y_rep_like  -----
##
## z test of coefficients:
##
##                     Estimate Std. Error z value Pr(>|z|)
## (Intercept)        -3.22561    0.26420  -12.21  < 2e-16 ***
## year2012            0.24164    0.01254   19.28  < 2e-16 ***
## year2016           -0.63298    0.04158  -15.22  < 2e-16 ***
## year2020            0.23755    0.03095    7.67  1.7e-14 ***
## year2024            0.40424    0.03165   12.77  < 2e-16 ***
## dem_therm          -0.01546    0.00211   -7.33  2.3e-13 ***
## rep_therm           0.02594    0.00126   20.61  < 2e-16 ***
## pid7Dem             0.13118    0.07376    1.78   0.0753 .
## pid7Rep             0.62231    0.05142   12.10  < 2e-16 ***
## party_diff2         0.59319    0.10226    5.80  6.6e-09 ***
## party_diff9         0.71698    0.09923    7.23  5.0e-13 ***
## party_more_consv2   0.47054    0.09690    4.86  1.2e-06 ***
## party_more_consv9   0.06354    0.11967    0.53   0.5955
## dem_ideoLib         0.68201    0.03973   17.17  < 2e-16 ***
## dem_ideoCon         0.13949    0.13114    1.06   0.2875
## rep_ideoLib         0.15641    0.12184    1.28   0.1993
## rep_ideoCon         0.64190    0.13185    4.87  1.1e-06 ***
## ideologyLib        -0.16791    0.05211   -3.22   0.0013 **
## ideologyCon         0.41304    0.05054    8.17  3.0e-16 ***
## gender2            -0.21013    0.03020   -6.96  3.4e-12 ***
## gender3           -10.81163    1.14359   -9.45  < 2e-16 ***
## educ7NoHS          -0.22672    0.03742   -6.06  1.4e-09 ***
## educ7BA_plus        0.49310    0.07997    6.17  7.0e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## -----  y_rep_dislike  -----
##
## z test of coefficients:
##
##                   Estimate Std. Error z value Pr(>|z|)
## (Intercept)       -0.84260    0.27576   -3.06  0.00225 **
## year2012          -0.46718    0.04453  -10.49  < 2e-16 ***
## year2016          -0.91663    0.05845  -15.68  < 2e-16 ***
## year2020          -0.60902    0.07114   -8.56  < 2e-16 ***
## year2024          -0.30706    0.07962   -3.86  0.00011 ***
## reg_census2        0.05473    0.03907    1.40  0.16122
## reg_census3        0.02020    0.05696    0.35  0.72292
## reg_census4        0.23984    0.05821    4.12  3.8e-05 ***
## dem_therm          0.00558    0.00205    2.72  0.00651 **
## rep_therm         -0.03150    0.00238  -13.25  < 2e-16 ***
## pid7Dem            0.35918    0.12061    2.98  0.00290 **
## pid7Rep            0.20709    0.06959    2.98  0.00292 **
```

```
## party_diff2        0.61140    0.12765    4.79  1.7e-06 ***
## party_diff9       -0.06093    0.53176   -0.11  0.90878
## party_more_consv2  0.60258    0.06156    9.79  < 2e-16 ***
## party_more_consv9  0.00978    0.07080    0.14  0.89016
## dem_ideoLib        0.57544    0.02327   24.73  < 2e-16 ***
## dem_ideoCon        0.34516    0.13521    2.55  0.01069 *
## rep_ideoLib        0.18391    0.01448   12.70  < 2e-16 ***
## rep_ideoCon        0.52611    0.03361   15.65  < 2e-16 ***
## ideologyLib        0.26717    0.05808    4.60  4.2e-06 ***
## ideologyCon        0.21834    0.06499    3.36  0.00078 ***
## age                0.01212    0.00306    3.96  7.5e-05 ***
## gender2           -0.29388    0.06146   -4.78  1.7e-06 ***
## gender3            1.82711    0.07187   25.42  < 2e-16 ***
## race_eth2         -0.21902    0.07748   -2.83  0.00470 **
## race_eth3         -0.50433    0.09713   -5.19  2.1e-07 ***
## race_eth4          0.18297    0.29834    0.61  0.53968
## race_eth5         -0.23165    0.08939   -2.59  0.00956 **
## race_eth6          0.05533    0.08420    0.66  0.51108
## educ7NoHS         -0.49781    0.05232   -9.51  < 2e-16 ***
## educ7BA_plus       0.61672    0.03827   16.12  < 2e-16 ***
## health_insur2     -0.14290    0.12851   -1.11  0.26617
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

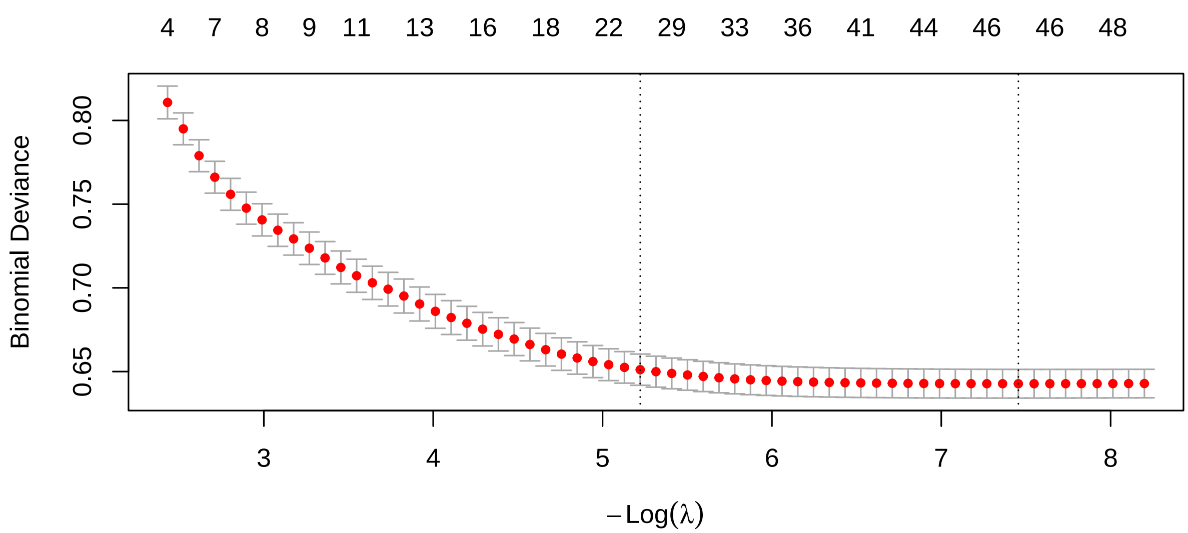
3 Robustness check: pooled outcome

A respondent is **in the open-ended sample** if they provided usable text on *any* of the four items. One binary outcome instead of four; if the same predictors fall out we have evidence the per-item models aren't picking up item-specific noise.

```
data_full$y_pooled <- as.integer(
  data_full$y_dem_like    == 1 |
  data_full$y_dem_dislike == 1 |
  data_full$y_rep_like    == 1 |
  data_full$y_rep_dislike == 1
)
table(data_full$y_pooled, useNA = "ifany")
```

```
##
##     0     1
##  2547 15297
```

```
Y_p   <- data_full$y_pooled
rhs_p <- setdiff(names(data_full), c(y_vars, "y_pooled"))
form_p <- as.formula(paste("~", paste(rhs_p, collapse = " + ")))
X_p   <- model.matrix(form_p, data = data_full)[, -1]
pf_p  <- ifelse(grepl("^year", colnames(X_p)), 0, 1)

set.seed(10)
fit.p.cv <- cv.glmnet(X_p, Y_p, alpha = 1, family = "binomial",
                      nfolds = 10, penalty.factor = pf_p)
plot(fit.p.cv)
```




```
coef.p <- extract.coef(fit.p.cv, lambda = fit.p.cv$lambda.1se)
coef.p
```

```
##                         Value SE       Coefficient
## (Intercept)       -1.55e+00 NA       (Intercept)
## year2012          -3.44e-01 NA          year2012
## year2016          -3.89e-01 NA          year2016
## year2020          -2.19e-01 NA          year2020
## year2024          -4.93e-02 NA          year2024
## reg_census4        1.01e-04 NA       reg_census4
## rep_therm         -5.97e-03 NA         rep_therm
## pid7Dem            7.42e-01 NA           pid7Dem
## pid7Rep            5.33e-01 NA           pid7Rep
## party_diff2        8.87e-01 NA       party_diff2
## party_more_consv2  4.12e-01 NA party_more_consv2
## party_more_consv9 -1.14e-01 NA party_more_consv9
## dem_ideoLib        4.66e-01 NA       dem_ideoLib
## dem_ideoCon        1.63e-01 NA       dem_ideoCon
## rep_ideoLib        6.03e-02 NA       rep_ideoLib
## rep_ideoCon        7.50e-01 NA       rep_ideoCon
## right_track1       7.98e-03 NA      right_track1
## right_track2      -1.30e-15 NA      right_track2
## ideologyLib        2.85e-01 NA       ideologyLib
## ideologyCon        5.77e-01 NA       ideologyCon
## age                1.99e-02 NA               age
## gender2           -2.04e-01 NA           gender2
## race_eth3         -1.46e-01 NA         race_eth3
## educ7NoHS         -1.08e-01 NA         educ7NoHS
## educ7BA_plus       4.54e-01 NA      educ7BA_plus
## health_insur2     -6.17e-02 NA     health_insur2
```

```r
vars.p <- parent_var(rownames(coef.p)[-1], names(data_full))
vars.p <- union(vars.p, "year")

f.p   <- as.formula(paste("y_pooled ~", paste(vars.p, collapse = " + ")))
fit.p <- glm(f.p, data = data_full, family = binomial)
V.p   <- sandwich::vcovCL(fit.p, cluster = ~ year, type = "HC1")
coeftest(fit.p, vcov. = V.p)
```

```
##
## z test of coefficients:
##
##                    Estimate Std. Error z value Pr(>|z|)
## (Intercept)       -2.170601   0.124657  -17.41  < 2e-16 ***
## year2012          -0.393330   0.039613   -9.93  < 2e-16 ***
## year2016          -0.459289   0.055673   -8.25  < 2e-16 ***
## year2020          -0.282811   0.059663   -4.74  2.1e-06 ***
## year2024          -0.117026   0.049758   -2.35   0.0187 *
## reg_census2       -0.009279   0.112467   -0.08   0.9342
## reg_census3        0.032893   0.066755    0.49   0.6222
## reg_census4        0.198306   0.080526    2.46   0.0138 *
## rep_therm         -0.010457   0.000949  -11.02  < 2e-16 ***
## pid7Dem            1.054400   0.084990   12.41  < 2e-16 ***
## pid7Rep            1.036033   0.104384    9.93  < 2e-16 ***
## party_diff2        0.899706   0.044634   20.16  < 2e-16 ***
## party_diff9        0.433706   0.157201    2.76   0.0058 **
## party_more_consv2  0.484896   0.060772    7.98  1.5e-15 ***
```

```
## party_more_consv9 -0.055983   0.131464   -0.43   0.6702
## dem_ideoLib        0.605467   0.096227    6.29  3.1e-10 ***
## dem_ideoCon        0.437113   0.065814    6.64  3.1e-11 ***
## rep_ideoLib        0.326032   0.112469    2.90   0.0037 **
## rep_ideoCon        0.907747   0.136430    6.65  2.9e-11 ***
## right_track2      -0.134509   0.060238   -2.23   0.0256 *
## ideologyLib        0.431635   0.077160    5.59  2.2e-08 ***
## ideologyCon        0.725487   0.047825   15.17  < 2e-16 ***
## age                0.024398   0.002421   10.08  < 2e-16 ***
## gender2           -0.336399   0.044708   -7.52  5.3e-14 ***
## gender3            0.771369   0.022950   33.61  < 2e-16 ***
## race_eth2          0.203289   0.107562    1.89   0.0588 .
## race_eth3         -0.492696   0.120606   -4.09  4.4e-05 ***
## race_eth4          0.337625   0.228120    1.48   0.1389
## race_eth5         -0.057278   0.102698   -0.56   0.5770
## race_eth6          0.119378   0.107483    1.11   0.2667
## educ7NoHS         -0.246377   0.176063   -1.40   0.1617
## educ7BA_plus       0.550000   0.068253    8.06  7.7e-16 ***
## health_insur2     -0.200354   0.121696   -1.65   0.0997 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

4 Summary

Across all four per-item models and the pooled robustness check, four predictor families do most of the work.

`y_dem_like` is the simplest model (17 predictors): partisanship, ideology placement, thermometers, and education explain who articulates Democratic likes.

`y_dem_dislike` is the richest model (23 predictors), pulling in age, race, gender, marital status, and work status. Articulating dislike of Democrats appears to be more demographically structured than articulating likes.

`y_rep_like` (15 predictors) is dominated by partisanship and ideology — demographics drop out almost entirely. Whether someone can articulate Republican likes is essentially a function of political engagement.

`y_rep_dislike` (20 predictors) again brings in demographics, similar to y_dem_dislike.

Below are some general findings -

1. Education is the single most consistent variable. `educ7BA_plus` is positive and highly significant in all four per-item models and the pooled model, and `educ7NoHS` is consistently negative. Holding everything else constant, college-educated respondents are roughly 60% more likely (in odds-ratio terms) to provide usable open-ended text than the high-school-or-some-college reference group.

2. Partisan strength matters more than direction. Both `pid7Dem` and `pid7Rep` show positive coefficients in most models, indicating that identifying with either party (versus pure independents, the reference) predicts response.

3. Political knowledge / engagement variables are extremely strong. `party_diff2` ("yes there's a difference between the parties") has beta = 0.55–0.92 in every model — one of the largest selected effects. `party_more_consv2` (correctly identifying Republicans as more conservative) is also consistently positive.

Beyond education, demographics matter most for the dislike outcomes:

1. Age is positive and significant for both dislike outcomes (beta = 0.0092-0.0121 per year): older respondents are more likely to articulate dislikes. It does not appear in either like model at lambda lse.

2. Gender is selected in three of four models. Women (`gender2`) are slightly less likely to give open-ended responses than men; the "other" category (`gender3`) shows large effects in opposite directions across items, but the small cell size warrants caution.

3. Race/ethnicity is selected for `y_dem_dislike` and `y_rep_dislike`. Asian (`race_eth3`) and Hispanic (`race_eth5`) respondents are consistently less likely to give open-ended responses than the white-non-Hispanic reference, all else equal.